

**Topic: AI-Based Mobile Money Fraud
Detection Using Machine Learning**

Table of Contents

Chapter 1: Introduction	4
1.1 Background of the Study.....	4
1.2 Problem Statement.....	4
1.3 Research Aim	4
1.4 Research Objectives.....	4
1.5 Research Questions	5
1.6 Research Significance.....	5
1.7 Study Scope.....	5
1.8 Dissertation Structure.....	5
Chapter 2: Literature Review	6
2.1 Introduction	6
2.2 Mobile Payment Fraud.....	6
2.3 Types of Mobile Payment Fraud	6
2.4 Machine Learning in Fraud Detection.....	6
2.5 Performance of Machine Learning Models.....	6
2.6 Explainable Artificial Intelligence (SHAP)	7
2.7 Past Studies.....	7
2.8 Research Gap	7
2.9 Conceptual Framework.....	7
2.10 Chapter Summary	7
Chapter 3: Research Methodology	8
3.1 Introduction and Overview of Methodology	8
3.2 Research Philosophy	9
3.3 Research Approach	9
3.4 Research Design.....	9
3.5 Research Strategy	9
3.6 Dataset Collection.....	9
3.7 Dataset Description	9
3.9 Exploratory Data Analysis	9
3.10 Data Preprocessing	9
3.11 Model Development	9
3.12 Model Evaluation	9
3.13 Explainability Analysis	9
3.14 Explainable AI Framework	9
3.15 Ethical Considerations.....	9
3.16 Limitations	9
3.17 Chapter Summary	10
Chapter 4: Data Analysis	11
4.1 Introduction	11

4.2 Exploratory Data Analysis	11
4.2.1 Dataset Overview	11
4.2.2 Dataset Features.....	11
4.2.3 Data Distribution	11
4.2.4 Correlation Analysis.....	11
4.2.5 Fraud Trend Analysis.....	11
4.2.6 Data Visualisation	11
4.3 Comparative Analysis of ML-Models	11
4.4 Secondary Thematic Analysis.....	12
4.5 Chapter Summary	12
Chapter 5: Results and Discussion	13
5.1 Introduction	13
5.2 Discussion and Interpretation of Results	13
5.3 Chapter Summary	13
Chapter 6: Conclusion and Recommendation	14
6.1 Conclusion.....	14
6.2 Linking with Aim and Research Questions.....	14
6.3 Research Limitation	14
6.4 Future Scope of Work	14
References:	15
Appendices:	17
Appendix-1: Summary of Papers	17

Chapter 1: Introduction

1.1 Background of the Study

The revolution of digital payment services and mobile connectivity has made banking and money transfer seamless. At the same time, it has also elevated the risk of payment/financial fraud in the digital payment system. A report by the BioCatch, based on data of 5 continents, has informed that 350 million consumers in the year 2025 have experienced financial fraud while transferring money through the digital payment services (BioCatch, 2026). The majority of such frauds were attributed to voice phishing scams, followed by romance scams and investment scams. SMS text-based phishing scams have also increased notably. In the Indian region only, between 2020 and 2021, the total value of financial frauds in the digital payment system rose by 20% in a single year (PWC, 2022). Such increased prevalence of fraud in digital payment systems has driven the need for advanced fraud detection and prevention measures.

1.2 Problem Statement

1.3 Research Aim

AIM:

The research aims to perform a comparative analysis of multiple Machine-Learning-based algorithms to build the most efficient and effective fraud detection model for the digital payment system and consumers.

1.4 Research Objectives

Objectives:

- To perform a literature review for establishing the theoretical aspect of the research in terms of the prevalence of mobile payment frauds, the drivers of payment frauds, performance & efficacy of ML-based algorithms and existing drawbacks of ML-based algorithms.
- To collect a dataset of mobile payment fraud using an online data collection platform, Kaggle.
- To perform exploratory data analysis for understanding the distribution of classes and trends in the dataset.
- To preprocess and transform the dataset based on the outcomes of the exploratory data analysis to make the dataset applicable for Machine Learning models.

- To compare the performance of multiple Machine Learning algorithms based on performance metrics like precision, accuracy, ROC-AUC curve and others
- To perform thematic analysis using secondary data to validate the outcomes of the experimental results and evaluate the causes of mobile payment frauds.

1.5 Research Questions

RQ-1: What are the ultimate root causes of mobile payment fraud, and how efficiently can a Machine Learning-based fraud detection model contribute to improving fraud detection and enhancing the mobile payment service?

Justification: The has limited understanding of the root causes of mobile payment fraud and is willing to understand the influence and impact of Machine Learning-based fraud detection model's contribution for improving the fraud detection while enhancing mobile payment service outcomes. By addressing the proposed research question, both qualitative (root causes of mobile payment frauds) and quantitative (Performance of the Machine Learning model) aspects of the research can be addressed.

1.6 Research Significance

1.7 Study Scope

1.8 Dissertation Structure

Chapter 2: Literature Review

2.1 Introduction

2.2 Mobile Payment Fraud

2.3 Types of Mobile Payment Fraud

2.4 Machine Learning in Fraud Detection

Over the past few years, several research studies have been organised to evaluate the performance of ML-based algorithms in fraud detection for payment services or financial services. However, the process often comes with several challenges. Compagnino et al., (2025) has performed a similar type of experiment and compared the performance of deep learning, supervised and non-supervised models. However, the primary issue experienced by the author is the "black-box" nature of the ML-models. Specifically, Deep-Learning-based algorithms have showcased this nature, which affects the decision interpretability of the detection and classification outcomes. This indicates that specific models affect the transparency of the decisions and classification, which limits the learning curve of experts in understanding the rationale behind the decisions. The findings by Vijayanand and Smrithy (2025) overcome this issue. As per the authors, the black box nature of the models can be eliminated by prioritising the application of Explainable AI (XAI) techniques such as SHAPs (SHapley Additive exPlanations). This AI technique enables the models to improve the transparency of the decisions by shedding light on the influential features and justifying the decisions. Therefore, the black-box nature can be neutralised with AI-based techniques.

2.5 Performance of Machine Learning Models

Researchers have also performed some comparative evaluation of the ML-based algorithms for determining the best-performing model in the context of mobile payment fraud detection. Lokanan (2023) has compared the performances of logistic regression, Gradient Descent, Decision Tree and Random Forest and has found that Random Forest remains the most consistent model for accurately determining fraud. In contrast, by comparing multiple tree-based approaches including Random Forest, Decision Tree

and others, Sengupta and Das (2023) has found that CatBoost remained a better-performing model for fraud detection and prevention compared to Random Forest. Hence, there is a knowledge gap in identifying and understanding the best algorithms for fraud detection.

2.6 Explainable Artificial Intelligence (SHAP)

2.7 Past Studies

2.8 Research Gap

2.9 Conceptual Framework

2.10 Chapter Summary

Chapter 3: Research Methodology

3.1 Introduction and Overview of Methodology

The project will perform a mixed methodology by following both the qualitative and quantitative techniques. Qualitative techniques are mainly applied for understanding human experience or behaviour, while quantitative methods are applied to analyse cause-and-effect relations between the variables (Xiong, 2022). Therefore, by following the qualitative methods, a secondary dataset will be collected to evaluate the root-causes and drivers of mobile payment frauds, whereas the quantitative method will be applied to perform the comparative analysis of the Machine Learning algorithms for determining the best model for fraud detection and prevention. In this quantitative approach, a dataset will be used for model training and testing purposes. Then, based on performance metrics like precision, accuracy and others, the comparative analysis of the models will be performed.

3.2 Research Philosophy

3.3 Research Approach

3.4 Research Design

3.5 Research Strategy

3.6 Dataset Collection

3.7 Dataset Description

3.9 Exploratory Data Analysis

3.10 Data Preprocessing

3.11 Model Development

3.12 Model Evaluation

3.13 Explainability Analysis

3.14 Explainable AI Framework

3.15 Ethical Considerations

3.16 Limitations

3.17 Chapter Summary

Chapter 4: Data Analysis

4.1 Introduction

4.2 Exploratory Data Analysis

4.2.1 Dataset Overview

<https://www.kaggle.com/datasets/denishazamuke/synthetic-mobile-money-transaction-dataset>

4.2.2 Dataset Features

Feature	Why useful?
step	Analyses the timing of transactions and fraud patterns over time.
type	Identifies which transaction types (e.g., TRANSFER, CASH_OUT) are more prone to fraud.
amount	Measures the transaction value, which is an important indicator of fraudulent activity.
oldbalanceOrg	Shows the sender's balance before the transaction and helps detect unusual behaviour.
newbalanceOrig	Compares the sender's balance after the transaction to identify suspicious changes.
oldbalanceDest	Shows the receiver's balance before the transaction.
newbalanceDest	Shows the receiver's balance after the transaction and helps identify abnormal balance changes.
isFraud	Target variable used to train and evaluate the Machine Learning models.
isFlaggedFraud	Indicates transactions automatically flagged as suspicious and can be used for comparison or additional analysis.

4.2.3 Data Distribution

4.2.4 Correlation Analysis

4.2.5 Fraud Trend Analysis

4.2.6 Data Visualisation

4.3 Comparative Analysis of ML-Models

4.3.1 Random Forest

4.3.2 Logistic Regression

4.3.3 Decision Tree

4.3.4 Gradient Descent

4.3.5 XG Boost

4.4 Secondary Thematic Analysis

4.5 Chapter Summary

Chapter 5: Results and Discussion

5.1 Introduction

5.2 Discussion and Interpretation of Results

5.2.1 Precision

- Measures how many predicted fraud transactions are actually fraud.

5.2.2 Accuracy

- Measures the overall percentage of correctly classified transactions.

5.2.3 Recall

- Measures how many actual fraud transactions are correctly detected.

5.2.4 F1-Score

- Balances Precision and Recall.
- Provides reliable evaluation for imbalanced fraud dataset

5.3 Chapter Summary

Chapter 6: Conclusion and Recommendation

6.1 Conclusion

6.2 Linking with Aim and Research Questions

6.3 Research Limitation

6.4 Future Scope of Work

Gann Chart

Research activities	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Topic Selection & Objectives												
Literature search												
Selecting methods												
Python Coding & ML Modeling												
Result & Analysis												
Concluding the research												
Final Draft Review & Submission												

CNN?

We will not use CNN because the PaySim dataset contains tabular transaction data, not image data. CNN is mainly designed for images and spatial data, so it will not be suitable for this dataset.

DNN?

If required, we will use DNN (Deep Neural Network) because it can learn complex patterns from tabular data and help identify fraudulent transactions. Otherwise, we will use tree-based models such as Random Forest and XGBoost, as they are expected to provide better performance for this type of fraud detection dataset because they handle tabular and highly imbalanced data more effectively.

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Appendices:

Appendix-1: Summary of Papers

Authors	Published Year	Aim	Methodology	Key Findings
Vijayanand and Smrithy	2025	To create financial fraud detection models that ensure higher accuracy and generate interpretable insights for stakeholders.	Experimental study to design Machine Learning models using explainable AI and a series of ML-based algorithms, including Logistic Regression, Decision Tree, Random Forest, XGBoost and others.	Decision tree remains the best-performing algorithm while showcasing better interpretability.
Hapase and Patil	2024	To design a telecommunication fraud-resilient framework for accurate detection of SMS phishing using AI-based techniques.	Experimental study using the CNN algorithm and advanced AI-based techniques such as Stemming, Tokenisation and others.	Compared to traditional fishing technique detection methods such as Naive Bayes, K-Nearest Neighbour and others, the

				Deep-Learning-based SMS fraud resilient model showcased higher Kappa statistics.
Zhdanova et al.,	2014	To propose an alternative method of ML-based Mobile Money Transfer service fraud detection by identifying money mules and fraudsters involved in the fraud chain.	Experimental study using the MMT simulator to compare the performance of multiple ML-based algorithms to propose the FCD method for ML detection based on PSA@R.	The FCD has demonstrated better precision and recall in fraud detection compared to Machine Learning algorithms.
Silva et al.,	2021	To design a multi-class supervised Machine-Learning-based model for financial fraud detection in mobile money service with adversarial auto-encoders.	Experimental study using Naïve Bayes, Random Forest and Support Vector Machines.	The models showcased better precision and accuracy when the latent vector dimensions generated through adversarial autoencoders

				decreased.
Hajek, Abedin and Sivarajah	2023	To perform a comparative evaluation of Machine Learning models for fraud detection in mobile payment systems.	Experimental study using 8 ML-based algorithms including KNN, Support Vector Machine, Random Forest, XGBoost, RUS + KNN, RUS + SVM, RUS + XGBoost, RUS+ RF, ABOD, CBLOF, HBOS, LODA, MCD and others.	RUS+ XGBoost remains the most efficient model because of the correct detection of the frauds and better cost efficiency compared to XGBOD and others.
Sengupta and Das	2023	To compare the performance of tree-based machine learning approaches for detecting financial fraud in mobile payment systems.	Experimental study using 7 tree-based approaches such as Decision Tree, Random Forest, GBM, AdaBoost, CGBost, CatBoost and LightGBM.	The best performing classifiers were CatBoost, Random Forest and Decision Tree, whereas the worst performing classifier was LightGBM.

Hussain and Asif	2020	To compare the performance of Machine-Learning-based classifiers for proposing the best method to detect anomalous transactions in mobile payment systems.	Experimental study using 8 ML-based classifiers including Decision Table, Ripper, Partial decision tree, J48, Decision Stump, Naive Bayes, Locally Weighted Learning and the K-Nearest Neighbour.	The best classifiers were RIPPER and J48 because of higher precision and recall, while Decision Stump and Naive Bayes remained the worst-performing models.
Lokanan	2023	To construct a fraud detection model using ML algorithms that can detect red flags of fraud and money laundering in mobile money transactions.	Experimental study using 4 ML-algorithms including Gradient Descent, Random Forest, Decision Tree and Logistic Regression.	Random Forest remains the most consistently performing model across all performance metrics when applied with sophisticated computation techniques rather than a rule-based approach.

Compagnino et al.,	2025	To examine different categories of financial frauds and analyse various ML approaches to evaluate and compare the performance of ML-based algorithms for financial fraud detection.	Comparative evaluation and analysis between supervised, unsupervised and deep learning approaches in a real-world bank transfer dataset.	Deep Learning approaches such as LSTM and GRU showcased better performance in capturing complex data patterns, but the black-box nature of the models becomes the biggest barrier to their applicability.
Thar and Wai	2025	To investigate and experiment with the combined implications of SMOTE and Feature Engineering hyper-parameter optimisation and improved detection of fraud in Mobile Money Services.	Experiment study using multiple ML-algorithms and ML-based techniques such as feature engineering, SMOTE and others.	The study informs that MSOTE and feature engineering improved the accuracy of traditional classifiers but cannot outperform the hybrid mode using HMM and XGBoost for consistency and accuracy

				of detection.
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